

MODULE 5 - CLIMATE ACTION (Part II) -Self-study Portion

GLOBAL CLIMATE AGREEMENTS

Global climate agreements are international treaties aimed at addressing climate change, with key examples including **the Paris Agreement, the Kyoto Protocol, and the United Nations Framework Convention on Climate Change (UNFCCC)**, all working towards limiting global warming and reducing greenhouse gas emissions.

THE UNITED NATIONS FRAMEWORK CONVENTION ON CLIMATE CHANGE

UNFCCC stands for United Nations Framework Convention on Climate Change. It was adopted on 5 June 1992. The objective of the UNFCCC is to stabilize greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system.

UNFCCC divides countries into two main groups:

Annex I countries - This group includes all the developed countries who have accepted GHG emissions reduction obligations and submit their annual greenhouse gas inventory.

Non-annex I countries - This group includes all the developing countries who have no GHG emissions reduction obligations, but may participate in the Clean Development Mechanism (CDM).

KYOTO PROTOCOL

The industrial revolution in the 19th century saw large-scale use of fossil fuels both for industrial activities and vehicular movements. In addition to the above, population explosion and urbanisation also have contributed to rise in greenhouse gases in the atmosphere. These greenhouse gases prevented heat, from our planet escaping into outer space, which resulted in global warming and changes in the climate all over the world.

To tackle the challenges posed by global warming and climate change, the United Nations Framework Convention on Climate Change (UNFCCC) initiated the framing of a protocol during the climate change meeting held at Kyoto, Japan in the year 1997. This later came to be known as the Kyoto Protocol. The Kyoto Protocol, adopted in Kyoto, Japan in December 1997, is an international agreement, setting limits on emission of greenhouse gases from industrialized countries. The Kyoto Protocol was adopted as the first addition to the UNFCCC. According to the UNFCCC website, the Protocol recognises that developed countries are principally responsible for the current high levels of GHG emissions in the atmosphere (as a result of more than 150 years of industrial activity) and places a heavier burden on developed nations. There are 192 parties to the convention: 191 states (including all the UN members

except Andorra, Canada, South Sudan and the United States) and the European Union. Kyoto Protocol has introduced the concept of Carbon Credit, Carbon Trading and Carbon foot print.

Greenhouse gases, such as carbon dioxide, (CO₂) methane (CH₄), nitrous oxide (N₂O) perfluorocarbons (PFCs), hydrofluorocarbons (HFCs), and sulphur hexa-fluoride (SF₆) affect the energy balance of the global atmosphere in ways expected to lead to an overall increase in global average temperature, known as global warming. The targets apply to the four greenhouse gases carbon dioxide, (CO₂) methane (CH₄), nitrous oxide (N₂O), (SF₆) and two groups of gases, sulphur hexa-fluoride, hydrofluorocarbons (HFCs) and Perfluorocarbons (PFCs). The six GHGs are translated into CO₂ equivalents, in determining the reduction in emissions.

PARIS AGREEMENT

Paris Summit - COP 21

Paris Climate Summit under the framework of United Nations Framework Convention on Climate Change (UNFCCC) in connection with Green House Gas (GHG) emission, mitigation and adaptation was attended by 195 countries which was held in December 2015 at Paris. The goal of this summit is to limit the global average increase in temperature below 2°C above pre-industrial level by the year 2100 and to continue efforts to further limit the temperature increase to 1.5°C. The countries including the global top ten GHG emitters (namely China, United States, European Union, India, Russia, Japan, Brazil, Indonesia, Mexico, and Iran) have pledged various GHG emission reduction measures. The UNFCCC warns that even if all the above nations stick on to their GHG emission reduction measures, still temperature rise would be around 3°C by the year 2100.

COP stands for Conference of Parties, an annual meeting of all nations that make up the United Nations Framework on Climate Change having 195 nations in total.

Aim of the Paris agreement

1. To achieve a legally binding agreement, with universal participation among all nations,
2. To keep global warming below what most scientists say is the critical threshold of 1.5 degrees Celsius of warming,
3. Making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development,
4. low greenhouse gas emissions development, in a manner that does not threaten food production,
5. Increasing the ability to adapt to the adverse impacts of climate change and foster climate resilience.

RENEWABLE ENERGY

Non-Conventional or Renewable energy is defined as energy that comes from resources which are naturally replaced on a human timescale, such as sunlight, wind, rain, tides, waves, and

geothermal heat. These are resources which are regenerated by natural processes, so that they can be used indefinitely. The advantages of these sources are that, they are free of pollution and are hence having very less impact on environment than the fossil fuels. The main drawback of these sources is that extraction of energy from these sources is expensive. Renewable energy replaces conventional fuels in four distinct areas such as electricity generation, air and water heating/cooling, automobile fuels and rural (off grid services). Few examples of non-conventional sources of energy are listed below:

- Solar Energy
- Wind power
- Hydro-electric power
- Biomass
- Ocean thermal power
- Bio-fuel
- Tidal power
- Ocean wave power
- Geothermal power

SOLAR ENERGY

The sun's rays or solar energy have been used since the beginning of time and is vital to all living things. In addition to solar energy being a constant source, heat and electricity are other forms of energy that can be made from solar energy. The largest advantage of solar energy is that it is a free and an unlimited source of energy.

Solar Energy Technologies

Solar energy can be captured in two forms-either as heat or electrical energy. There are different methods of harnessing solar radiations so that it can be harnessed into usable energy. These methods use either active solar energy or passive solar energy. Active methods use devices for converting solar energy into other forms of energy such as heat or electricity. Active methods include use of photovoltaics and concentrated solar power (CSP). Passive solar techniques do not use any devices, instead they make use of the climate or the natural processes. Passive techniques include orientation of the building to the sun, select materials with favourable properties, and by designing open spaces for natural air circulation. There are a number of solar techniques which provides a broad number of applications, of which some of them are listed below,

1. Solar water heating
2. Solar space heating of buildings
3. Solar air conditioning
4. Solar drying
5. Solar green-houses
6. Solar desalination
7. Solar electricity - Thermal

8. Solar refrigeration
9. Solar cooking
10. Solar furnaces
11. Salt production
12. Solar electricity - Photovoltaic

Advantages of Solar Energy

- Solar energy is free
- Solar energy does not cause pollution
- Solar energy can be used in remote areas where it is too expensive to extend the electricity power grid
- Many everyday items such as calculators and other low power consuming devices can be powered by solar energy effectively
- It is estimated that the world's oil reserves will last for 40 to 50 years. On the other hand, solar energy is infinite (forever).

Disadvantages of Solar Energy

- Solar energy can only be harnessed when it is day time and sunny
- Solar collectors, panels and cells are relatively expensive to manufacture although prices are falling rapidly
- Solar power stations can be built but they do not match the power output of similar sized conventional power stations. They are also very expensive too.
- Large areas of land are required to capture the sun's energy
- Solar power can be used to charge batteries (so that solar powered devices can be used at night)

BIOMASS

Biomass is any organic material which has stored sunlight in the form of chemical energy (by the process of photosynthesis, whereby the plants use chlorophyll to convert the energy in the sun's rays into stored energy in the plants). As a fuel it may include wood, wood waste, straw, manure, sugarcane, and many other by-products from a variety of agricultural processes. They can be burnt to produce energy (in the form of heat, electricity or liquid or gaseous biofuels). India is currently the world's largest biomass energy producer as there are around 800 million Indians who use some form of biomass as their primary or only energy source for cooking.

Biomass falls into three categories:

- Biomass in its traditional form (Wood and Agricultural residue is burnt to produce energy)
- Biomass in its non-traditional form (Here biomass is converted to ethyl alcohol and methyl alcohol to be used as liquid fuels in engines)

- Biomass for domestic use (Here the organic matter in waste is decomposed anaerobically to produce a mixture of gases (called biogas) namely methane, carbon-dioxide, hydrogen sulphide etc. Biogas is a good biofuel used for cooking and lighting)

Examples of Biomass

- Wood - Trees, Shrubs, Wood residue, Sawdust, Bark, etc.
- Wastes - Municipal Solid Waste (MSW), Paper, Food wastes etc.
- Crops
 1. Starch crops - Corn, Wheat, and Barley
 2. Sugar crops - Cane and Beet
 3. Oil-seed crops - Soyabean and Sunflower
 - Aquatic Plants - Algae, Water weed, Water hyacinth etc.

Methods of converting biomass to energy are:

- Burning
- Alcohol fermentation
- Pyrolysis
- Anaerobic digestion

Advantages

- Theoretically inexhaustible fuel source
- When direct combustion of plant mass is not used to generate energy (i.e. fermentation, pyrolysis, etc. are used instead), there is minimal environmental impact
- Alcohols and other fuels produced by biomass are efficient, viable, and relatively clean-burning.

Disadvantages

- Could contribute a great deal to global warming and particulate pollution if directly burnt
- Still an expensive source, both in terms of producing the biomass and converting it to alcohols.

WIND ENERGY

Wind is the result of pressure variations experienced on the earth's surface due to unequal heating by the sun. It blows from a region of high pressure to that of lower one. Wind mills are used for harnessing these air flows and wind turbines are used to produce power. Areas where winds are stronger and more constant, such as offshore and high-altitude sites, are preferred locations for wind farms. Globally, the long-term technical potential of wind energy is believed to be five times the total current global energy production, or 40 times current electricity demand, assuming that all practical barriers are overcome. This would require wind turbines to be installed over large areas, particularly in areas of higher wind resources, such as offshore.

Wind power is one of the world's fastest expanding sources of energy. Wind energy is a renewable source of energy as well as a non-polluting one. Although these are environmentally friendly means of energy, they take away large open spaces. They can also hinder the path of migratory winds.

Advantages

- Wind is free, wind farms need no fuel
- Produces no waste or greenhouse gases
- The land beneath the wind mill can be used for farming
- A good method of supplying energy to remote areas

Disadvantages

- High initial investment required
- Wind is not always predictable, hence do not produce the same amount of power at all times
- Can kill birds - migrating flocks tend to like strong winds. However, this is rare, as we tend not to build wind farms on migratory routes anyway
- Noise produced by the rotor blades is a concern for the local people residing near wind turbines.

NEW ENERGY SOURCES

HYDROPOWER

Hydro-electric Power: Hydroelectricity refers to the electricity generated by hydropower; the production of electric power through the use of gravitational force of the falling or flowing water. The name comes from "hydro", the Greek word for water. Hydropower is a renewable, clean and non-polluting energy resource. A dam is a hydraulic structure constructed across a flowing river to store water on its upstream side and the artificial water body created as a result of construction of the dam on the upstream side is called the storage reservoir. Gravitational potential energy is stored in the water the upstream of the dam. Water flows from higher elevation to lower elevation through tunnels in the dam, to turn turbines. Because of the great height of the water, it will arrive at the turbines at high pressure and make use of the kinetic energy in running water to turn turbines to generate electrical power. It is to be noted that the dam is much thicker at the bottom than at the top, because the pressure of the water increases with depth.

the largest of which is the Three Gorges Dam in China and a smaller one is the Akosombo Dam in Ghana Micro hydro systems are hydroelectric power installations that typically produce up to 100 kW of power. They are often used in water rich areas as a Remote-Aren Power Supply (RAPS).

GEOTHERMAL

The name "geothermal" comes from two Greek words: "geo" means "Earth" and "thermal" means "heat". It is the energy stored in the form of heat below the earth's surface. Its potential is limitless and is a clean and sustainable source. This requires that the hot rock be relatively shallow, so it is site-specific and can only be applied in geologically active areas. The centre of the Earth is around 5000°C easily hot enough to melt rock. In general, the temperature rises by 1°C for every 30 - 50 metres we go down, but this does vary depending on location. In volcanic areas, molten rock can be very close to the surface. Geothermal energy is an important resource in volcanically active places such as Iceland and New Zealand. Geothermal Energy can be captured by two methods:

- Geothermal power plants
- Geothermal heat pumps

Advantages

- Inexhaustible and renewable energy source
- Non-polluting and eco-friendly source
- An excellent supplement to other renewable sources
- Geothermal energy is not affected by seasonal changes.

Disadvantages

- Not available in many locations
- Overall efficiency of power production is quite low (10-20%)

ENERGY DERIVED FROM OCEANS

Ocean energy has vast potential as 70% of the earth is covered with water. The tides that hit sea shore have enormous potential in them and can be used to convert to electrical energy. Ocean energy can be captured via three ways:

(a) Tidal Energy

(b) Wave Energy

(c) OTEC (Ocean Thermal Energy Conversion).

(a) Tidal energy captures kinetic energy from incoming and outgoing tides and tidal turbines uses that kinetic energy and convert it to electrical energy.

Advantages of Tidal Energy

- Tidal power is free from pollution
- It is one of the cleanest sources of renewable energy
- It can provide recreational facilities to tourists
- Low noise pollution.

Disadvantages of Tidal Energy

- A dam constructed across an estuary is very expensive to build, and affects fish and marine ecosystem
- There are only a few suitable sites for generating tidal power
- The supply of power is not continuous as it depends upon the timing of tides.

(b) Wave energy is captured directly from the surface of the waves.

Advantages

- The energy generated is free and environment friendly
- Abundant and widely available
- Not expensive to operate and maintain.

Disadvantages

- Depends on the waves sometimes we'll get lots of energy, sometimes almost nothing (i.e. energy output is variable)
- Needs a suitable site, where waves are consistently strong
- Components of a wave power station should withstand

(c) OTEC produces electricity by using the temperature difference between deep cold ocean water and warm tropical surface water.

Advantages

- Clean and renewable energy source
- Power from OTEC is continuous

Disadvantages

- This technology is relatively inefficient
- Is suitable only for tropical seas with relatively large temperature gradients